DATASHEET

MATRIX
CIRCUIT BOARD MATERIALS

# KB-6167F High Performance Laminates

KB
KINGBOARD

## Benefits

- Able to withstand multiple thermal excursions during PCB fabrication and assembly
- High Thermal Resistance and Long-Term Reliability
- Wide operating window for multilayer processing
- Excellent Thermal Shock Reliability
- Withstands Thermal Cycling, IST and other accelerated testing methods
- UV blocking for maximum compatibility with automated optical inspection
- Excellent Dimensional Stability
- RoHS/REACH Compliant

## Applications

- High Layer Count PCBs
- Backplanes
- High Complexity Multilayers
- Industrial Electronics
- Commercial Electronics
- High-End Servers
- Wireless Communication Infrastructure
- Automotive Applications Requiring High Thermal Resistance

## High Reliability, High Tg, FR-4 Multilayer Material

KB-6167F is specifically formulated to withstand increasingly stringent demands of high complexity, high layer, lead-free PCB designs and applications.

This high Tg, multi-functional phenolic cured resin system with inorganic fillers result in high performing PCB structures.

KB-6167F offers very high heat resistance. It is particularly well-suited for lead-free soldering processes.

## Features

- High Tg (DSC) value of 170-180°C
- Low Moisture Absorption
- CAF-Enhanced
- High Decomposition Temperature: > 340°C
- Lead-Free Assembly Compatible
- Low Z-Axis Expansion

## Laminate Properties

| Item | | Typical Value | Test Method (IPC-TM-650) | Test Condition | | Unit | Specification |
|---|---|---|---|---|---|---|---|
| Thermal | Thermal Stress | ≥ 240 | 2.4.13.1 | Float 288°C/ unetched | | sec | ≥ 10 |
| | Glass Transition (Tg) | 175 | 2.4.25 | E-2/105 DSC | | °C | ≥ 170 |
| | CTE/Z-Axis Expansion | 40 | 2.4.24 | Alpha 1 | | ppm/°C | ≤ 60 |
| | | 230 | | Alpha 2 | | | ≤ 300 |
| | | 2.6 | | 50 - 260°C | | % | ≤ 3.0 |
| | X/Y CTE | 12/15 | 2.4.24 | 40 - 125°C | | ppm/°C | — |
| | T-260 | > 60 | 2.4.24.1 | TMA | | min | ≥ 30 |
| | T-288 | > 35 | 2.4.24.1 | TMA | | min | ≥ 15 |
| | TD (5% weight loss) | 349 | 2.4.24.6 | TGA | | °C | > 340 |
| | Flammability | V-0 | UL94 | E-24/23 | | Rating | V-0 |
| Electrical | Surface Resistivity | ≥ $10^9$ | 2.5.17.1 | C-96/35/90 | | MΩ | ≥ $10^4$ |
| | Volume Resistivity | ≥ $10^{10}$ | 2.5.17.1 | C-96/35/90 | | MΩ-cm | ≥ $10^6$ |
| | Dielectric Breakdown | 67 | 2.5.6 | D-48/50+D0.5/23 | | kV | ≥ 40 |
| | Dielectric Constant | 4.8 | 2.5.5.2 | Etched (R/C 50%) | @ 1MHz | — | ≤ 5.4 |
| | | 4.6 | | | @ 1GHz | | |
| | Loss Tangent | 0.015 | 2.5.5.2 | Etched (R/C 50%) | @ 1MHz | — | ≤ 0.035 |
| | | 0.016 | | | @ 1GHz | | |
| | CTI | > 175 | IEC60112 | A | | V | — |
| | Arc Resistance | 129 | 2.5.1 | D-48/50+D0.5/23 | | sec | ≥ 60 |
| Mechanical | Peel Strength ( 1oz) | 1.2 | 2.4.8 | 125°C | | N/mm | ≥ 0.70 |
| | | 1.3 | | Float 288°C/10 sec | | | ≥ 1.05 |
| | | 1.1 | | After Process Solution | | | ≥ 0.80 |
| | Flexural Strength | 540 | 2.4.4 | Length Direction | | $N/mm^2$ | ≥ 415 |
| | | 480 | | Cross Direction | | | ≥ 345 |
| | Moisture Absorption | 0.09 | 2.6.2.1 | D-24/23 | | % | ≤ 0.5 |

Remarks:
- Typical values for reference only.
- Standard values according to IPC-4101E/126
- Typical value of specimen thickness is 1.6mm (8 * 7628)

UL: E123995 RoHS

DATASHEET

# KB-6167F High Performance Laminates

## KB-6167 Laminate Constructions

| Thickness (mil) | Layup | RC (%) | Dk | | | | Df | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | 1G | 2G | 5G | 10G | 1G | 2G | 5G | 10G |
| 2.0 | 106x1 | 73 | 4.1 | 4.0 | 3.9 | 3.9 | 0.017 | 0.017 | 0.018 | 0.019 |
| 2.5 | 1067x1 | 71 | 4.1 | 4.1 | 4.0 | 3.9 | 0.017 | 0.017 | 0.018 | 0.019 |
| 3.0 | 1080x1 | 67 | 4.2 | 4.1 | 4.1 | 4.0 | 0.016 | 0.016 | 0.017 | 0.018 |
| 3.5 | 3313x1 | 52 | 4.5 | 4.4 | 4.4 | 4.3 | 0.015 | 0.015 | 0.016 | 0.017 |
| 4.0 | 2116x1 | 47 | 4.6 | 4.5 | 4.5 | 4.4 | 0.014 | 0.015 | 0.016 | 0.017 |
| 4.5 | 2116x1 | 51 | 4.5 | 4.5 | 4.4 | 4.3 | 0.015 | 0.015 | 0.016 | 0.017 |
| 5.0 | 2165x1 | 49 | 4.6 | 4.5 | 4.4 | 4.4 | 0.015 | 0.015 | 0.016 | 0.017 |
| 5.5 | 1056x1 | 42 | 4.7 | 4.6 | 4.6 | 4.5 | 0.014 | 0.014 | 0.015 | 0.016 |
| 6.0 | 1506x1 | 46 | 4.6 | 4.6 | 4.5 | 4.4 | 0.014 | 0.014 | 0.015 | 0.016 |
| 7.5 | 7628x1 | 45 | 4.6 | 4.6 | 4.5 | 4.5 | 0.014 | 0.014 | 0.015 | 0.016 |
| 8.0 | 7628x1 | 48 | 4.6 | 45 | 4.4 | 4.4 | 0.015 | 0.015 | 0.016 | 0.017 |
| 10.0 | 2165x2 | 50 | 4.5 | 4.5 | 4.4 | 4.4 | 0.015 | 0.015 | 0.016 | 0.017 |
| 12.0 | 1506x2 | 42 | 4.7 | 4.6 | 4.6 | 4.5 | 0.014 | 0.014 | 0.015 | 0.016 |
| 15.0 | 7628x2 | 45 | 4.6 | 4.6 | 4.5 | 4.5 | 0.014 | 0.014 | 0.015 | 0.016 |
| 18.0 | 7628x2+1080x1 | 47 | 4.6 | 4.5 | 4.5 | 4.4 | 0.014 | 0.015 | 0.016 | 0.017 |
| 21.0 | 7628x3 | 42 | 4.7 | 4.6 | 4.6 | 4.5 | 0.014 | 0.014 | 0.015 | 0.016 |
| 25.0 | 7628x3 | 50 | 4.5 | 4.5 | 4.4 | 4.4 | 0.015 | 0.015 | 0.016 | 0.017 |
| 30.0 | 7628x4 | 45 | 4.6 | 4.6 | 4.5 | 4.5 | 0.014 | 0.014 | 0.015 | 0.016 |
| 35.0 | 7628x5 | 42 | 4.7 | 4.6 | 4.6 | 4.5 | 0.014 | 0.014 | 0.015 | 0.016 |
| 40.0 | 7628x5 | 48 | 4.6 | 4.5 | 4.4 | 4.4 | 0.015 | 0.015 | 0.016 | 0.017 |
| 47.0 | 7628x6 | 44 | 4.7 | 4.6 | 4.5 | 4.5 | 0.014 | 0.014 | 0.015 | 0.016 |
| 59.0 | 7628x8 | 42 | 4.7 | 4.6 | 4.6 | 4.5 | 0.014 | 0.014 | 0.015 | 0.016 |

## KB-6067F Prepreg Specifications

| Glass Style | RC (%) | Thickness (mil) | Dk | | | | Df | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | 1G | 2G | 5G | 10G | 1G | 2G | 5G | 10G |
| 106 | 71 | 1.97 | 4.1 | 4.1 | 4.0 | 3.9 | 0.017 | 0.017 | 0.018 | 0.019 |
| | 73 | 2.14 | 4.1 | 4.0 | 3.9 | 3.9 | 0.017 | 0.017 | 0.018 | 0.019 |
| 1080 | 62 | 2.75 | 4.3 | 4.2 | 4.2 | 4.1 | 0.016 | 0.016 | 0.017 | 0.018 |
| | 65 | 3.03 | 4.2 | 4.2 | 4.1 | 4.0 | 0.016 | 0.016 | 0.017 | 0.018 |
| | 68 | 3.37 | 4.2 | 4.1 | 4.0 | 4.0 | 0.016 | 0.017 | 0.018 | 0.019 |
| | 70 | 3.58 | 4.1 | 4.1 | 4.0 | 3.9 | 0.017 | 0.017 | 0.018 | 0.019 |
| 3313 | 52 | 3.5 | 4.5 | 4.4 | 4.4 | 4.3 | 0.015 | 0.015 | 0.016 | 0.017 |
| | 55 | 3.75 | 4.4 | 4.4 | 4.3 | 4.3 | 0.015 | 0.015 | 0.016 | 0.017 |
| | 58 | 4.05 | 4.4 | 4.3 | 4.2 | 4.2 | 0.016 | 0.016 | 0.017 | 0.018 |
| | 50 | 4.27 | 4.5 | 4.5 | 4.4 | 4.4 | 0.015 | 0.015 | 0.016 | 0.017 |
| 2116 | 52 | 4.49 | 4.5 | 4.4 | 4.4 | 4.3 | 0.015 | 0.015 | 0.016 | 0.017 |
| | 54 | 4.74 | 4.5 | 4.4 | 4.3 | 4.3 | 0.015 | 0.015 | 0.016 | 0.017 |
| | 56 | 5.02 | 4.4 | 4.4 | 4.3 | 4.2 | 0.015 | 0.015 | 0.016 | 0.017 |
| | 58 | 5.31 | 4.4 | 4.3 | 4.2 | 4.2 | 0.016 | 0.016 | 0.017 | 0.018 |
| 1506 | 44 | 5.78 | 4.7 | 4.6 | 4.5 | 4.5 | 0.014 | 0.014 | 0.015 | 0.016 |
| | 46 | 6.06 | 4.6 | 4.6 | 4.5 | 4.4 | 0.014 | 0.014 | 0.015 | 0.016 |
| | 48 | 6.36 | 4.6 | 4.5 | 4.4 | 4.4 | 0.015 | 0.015 | 0.016 | 0.017 |
| 7628 | 42 | 7.03 | 4.7 | 4.6 | 4.6 | 4.5 | 0.014 | 0.014 | 0.015 | 0.016 |
| | 44 | 7.34 | 4.7 | 4.6 | 4.5 | 4.5 | 0.014 | 0.014 | 0.015 | 0.016 |
| | 47 | 7.90 | 4.6 | 4.5 | 4.5 | 4.4 | 0.014 | 0.015 | 0.016 | 0.017 |
| | 50 | 8.52 | 4.5 | 4.5 | 4.4 | 4.4 | 0.015 | 0.015 | 0.016 | 0.017 |
| 7630 | 48 | 8.48 | 4.6 | 4.5 | 4.4 | 4.4 | 0.015 | 0.015 | 0.016 | 0.017 |
| | 50 | 8.93 | 4.5 | 4.5 | 4.4 | 4.4 | 0.015 | 0.015 | 0.016 | 0.017 |

Standard Sizes: Panel yields based on 48"x42", 48"x36" Sheets (Other sizes may be available by special request)
Copper Cladding: Hoz, 1oz, 2oz, 3oz (Heavier copper may be available by special request)

MATRIX
CIRCUIT BOARD MATERIALS

Supplier to the North American Circuit Board Industry since 1977


**TORONTO**
1124 Mid-Way Boulevard,
Mississauga, ON L5T 2C1
Phone (905) 670-8400
Fax (905) 670-8420

**SANTA ANA**
2730 S. Main Street,
Santa Ana, CA 92707
Phone (714) 825-0404
Fax (714) 825-0406

**SANTA CLARA**
3261 Edward Avenue,
Santa Clara, CA 95054
Phone (408) 477-2963
Fax (408) 564-7047

**CHICAGO**
770 Creel Drive,
Wood Dale, IL 60191
Phone (630) 475-8753

**Toll-free 1-800-668-5447**

Visit us on the web at **www.matrixusa.us**



The information in this document is believed to be accurate, but Matrix makes no implied or expressed warranties to that accuracy and assumes no liability arising from its use. Users should conduct their own tests to determine the suitability of these products for their particular application. The listed data is within the normal range of product properties, but should not be the sole criteria for application design.

Copyright © Matrix USA Inc. All rights reserved. Rev 20200616